

Dinosaur Recalibration

A Lognormal Anatomy-Paradigm Sensitivity Model for Crowd-Sourced Bradley–Terry Combat Scores

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Abstract

This note fits Bradley–Terry (BT) power ratings to approximately 4,494 crowd-sourced animal-combat matchup entries and asks how the resulting ranking changes under an explicit uncertainty model for a hypothesized dinosaur “anatomy advantage.” The motivating claim is not that the crowd data themselves identify such an advantage. Rather, the crowd scale is treated as a Cenozoic-calibrated baseline and the anatomy correction as an elicited sensitivity parameter. Two calibrations define the elicitation: a deliberately conservative 3.18× correction obtained by log-midpoint shrinkage of a stronger Dakotaraptor calibration, and a 7.27× direct Utahraptor calibration. The working model is

$$\log F_{\text{anat}} \sim \mathcal{N}(1.43251, 0.25091^2),$$

whose median is 4.189×. The log-space standard deviation corresponds to a multiplicative one-sigma factor of 1.285; hence the one- and two-log-sigma bands are 3.26–5.38× and 2.54–6.92×, respectively. The original 3.18–7.27 interval is used only to elicit these parameters and is not treated as a confidence interval thereafter. At the median correction, a raw Velociraptor score of about 1.66 becomes about 6.9, Deinonychus about 126, Dakotaraptor about 801, Utahraptor about 1,118, and Tyrannosaurus about 44,000 on the Lion = 100 scale. The exercise remains a transparent sensitivity analysis, not an empirical estimate of dinosaur combat performance, but the probabilistic formulation makes the anatomical hypothesis explicit, graded, and falsifiable in principle.

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1 Introduction

The interactive essay that motivated this note contrasts two ways of intuitively ranking animal formidability: a crowd “weight paradigm,” in which larger animals tend to be judged stronger, and an “anatomy paradigm,” in which weapon geometry, posture, locomotion, leverage, armor, and attack mechanics can move animals far from a mass-only expectation [1]. The broader interpretive hypothesis is that modern intuitions are calibrated entirely inside a post-K–Pg, Cenozoic ecology. If the disappearance of non-avian dinosaurs constituted a major ecological release for mammals, then an observer raised among Cenozoic body plans may systematically underestimate what strongly weaponized Mesozoic animals would have been like in practice.

That motivating idea is deliberately stronger than what the present data can prove. A community-voting dataset cannot establish that “Cenozoic paradigm-blindness” exists, much less measure an ecosystem-wide phase transition in selection pressure. What it can do is provide a baseline ranking on which an alternative calibration can be made explicit. This paper therefore asks a narrower question:

If the crowd ranking is the baseline and the proposed anatomy correction is uncertain rather than fixed, what does the ranking look like under a transparent probability model for that correction?

The central change from the earlier version of this note is accordingly methodological. Instead of multiplying all dinosaurs by a single deterministic factor, the anatomy multiplier is modeled as a lognormal random variable. Its median and dispersion are elicited from two conservative calibration constructions. Once those parameters are fixed, all downstream tables use the median correction and standard-deviation bands; the original elicitation interval drops out of the substantive analysis.

The paper remains an exploratory technical companion to the public visualization, not a zoological measurement paper. Its purpose is to make a provocative intuition inspectable rather than to disguise it as a result of the crowd data.

2 Data

2.1 Source

The raw data consists of pairwise animal-combat matchups scraped from the Carnivora.net Interspecific Conflict Directory [8]. Each matchup records two animals, a win probability for the first animal as inferred from site-user voting, and the number of votes. After scraping, the working dataset contains approximately 4,494 matchup entries spanning approximately 2,369 unique animals.

2.2 Unicode normalization

The source data contains corrupted Unicode characters (U+FFFD replacement characters from encoding errors). We apply an allowlist-based cleaning procedure: retain only ASCII letters, digits, and common punctuation; replace corrupted characters with a space at word boundaries and delete them within words. A manual corrections dictionary handles remaining ambiguous cases.

2.3 Filtering

Matchups with reported win probabilities below 5% or above 95% are excluded from the fitted graph. This is a stability choice, not a claim that near-unanimous matchups contain no information. Extreme proportions can create near-separation and dominate a finite-score BT fit when vote counts and graph connectivity are uneven. No threshold-sensitivity analysis is reported here, so conclusions that depend strongly on the truncation should not be treated as robust.

3 Graph Preprocessing

The raw matchup graph contains disconnected components and weakly anchored entries. In a Bradley–Terry model, disconnected components do not possess a data-determined relative location on a common score scale. The preprocessing pipeline therefore removes some isolated structures and adds explicit pseudo-matchups to regularize others. These pseudo-observations are modeling assumptions, not independent biological measurements.

3.1 Graph cleanup

We exclude three classes of poorly connected animals: (i) animals whose only matchups have 0% or 100% reported win probability (283 animals); (ii) two-animal islands in the real graph (228 animals across 114 pairs); and (iii) larger islands with no plausible bridge to the main component (39 animals across 11 isolated clusters). This removes 550 animals. The remaining 1,819 animals and 4,087 retained real matchups form the basis for structural augmentation.

3.2 Gender constraints

For animals with male, female, and base entries, we add two pseudo-matchups designed to pull the base score toward the geometric mean of the male and female scores. If both pseudo-matchups use the same win probability p , then in isolation the construction gives

$$\pi_{\text{base}}^2 = \pi_{\text{male}}\pi_{\text{female}}.$$

We use $p = 0.60$ at pseudo-weight 10. In the full graph these constraints regularize rather than force equality.

3.3 Group-to-solo scaling

Entries such as “Grey Wolf (pack of 5)” are linked to the solo entry with $P = N^{0.7}/(N^{0.7} + 1)$ at pseudo-weight 3. The exponent 0.7 is a heuristic for diminishing returns from imperfect coordination; it is not estimated from the Carnivora data.

3.4 Island bridges and duplicate linking

Remaining disconnected clusters receive 12 manual bridge pseudo-matchups at weight 3. Generic/regional duplicates receive 50% links at weight 20, while subspecies links use size-adjusted probabilities at weight 10. These constraints repair identifiability but necessarily make the placement of weakly connected animals conditional on the augmentation design.

3.5 Summary

The preprocessing adds 388 synthetic matchups to 4,087 retained real matchups, producing a connected augmented graph of 1,819 animals. The exported taxonomy counts currently sum to 1,812 rather than 1,819; this seven-entry discrepancy is a bookkeeping issue in the classification export and is retained explicitly rather than silently reconciled.

4 Bradley–Terry Model

Each animal i receives a latent strength $\pi_i > 0$ with

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j}. \quad (1)$$

Writing $\theta_i = \log \pi_i$, we maximize a weighted BT log-likelihood with L_2 penalty $\lambda = 0.01$. Real matchup vote totals supply fractional win/loss counts; pseudo-matchups use their assigned pseudo-weights. Scores are normalized after fitting so that Lion = 100.

The BT score should be interpreted as a one-dimensional summary of this particular voting graph. Real combat is context-dependent, and a one-dimensional latent ordering cannot encode all matchup-specific interactions.

5 Taxonomy and Scaling Classes

Animals are assigned to three scaling categories:

1. **Baseline / unscaled:** extant animals, Cenozoic and Paleozoic extinct species, synapsids, and Mesozoic mammals.
2. **Dinosaur:** non-avian Dinosauria from the Mesozoic.
3. **Mesozoic-other:** pterosaurs, marine reptiles, Mesozoic crocodylomorphs, and other non-dinosaurian Mesozoic reptiles represented in the dataset.

The first category is intentionally called baseline rather than “non-Mesozoic” because it includes Mesozoic mammals.

6 The Probabilistic Anatomy Paradigm

6.1 Calibration ladder

All scores use Lion = 100, and the fitted Human score is 5.15. The earlier strong calibration placed Dakotaraptor above Lion by the same multiplicative ratio by which Lion lies above Human:

$$\frac{D_{\text{target}}}{100} = \frac{100}{5.15}, \quad D_{\text{target}} = 1,940.$$

Against Dakotaraptor’s raw crowd score of about 191.3, this gives a direct factor of approximately 10.14×

That value is retained only as a motivating extreme calibration. A single conservative log-space shrinkage toward the uncorrected crowd scale gives

$$L = \sqrt{10.14} = 3.18.$$

As a second calibration, use Utahraptor rather than Dakotaraptor in the same Lion-as-geometric-mean construction. The previous export gives Utahraptor a raw crowd score of approximately 2,705/10.14 = 266.8. Requiring its target score to be 1,940 therefore gives

$$U = \frac{1,940}{266.8} \approx 7.27.$$

The two numbers $L = 3.18$ and $U = 7.27$ are not declared confidence bounds. They are elicitation anchors: one deliberately shrunken and one direct but based on the more conservative Utahraptor comparison.

6.2 Lognormal uncertainty model

The anatomy factor F_{anat} is modeled in log space. Because the lower anchor has already undergone an explicit conservative shrinkage while the upper anchor is an unshrunk direct calibration, the center is intentionally pulled two-to-one toward the lower value:

$$\mu = \frac{2}{3} \log 3.18 + \frac{1}{3} \log 7.27 = 1.43251. \quad (2)$$

Thus the median factor is

$$\tilde{F}_{\text{anat}} = e^\mu = 3.18^{2/3} 7.27^{1/3} = 4.189. \quad (3)$$

The log-space standard deviation is then chosen so that the total probability outside the two elicitation anchors is 15%:

$$\Phi\left(\frac{\log 3.18 - \mu}{\sigma}\right) + 1 - \Phi\left(\frac{\log 7.27 - \mu}{\sigma}\right) = 0.15. \quad (4)$$

Numerically,

$$\log F_{\text{anat}} \sim \mathcal{N}(1.43251, 0.25091^2). \quad (5)$$

Because the center is closer to the lower anchor, the two elicitation tails are asymmetric: approximately 13.6% lies below 3.18 and 1.4% above 7.27. This asymmetry is intentional. Once Eq. (5) has been specified, however, the original anchors are no longer used as an interpretive interval; downstream analysis is stated directly in terms of the median and standard deviations of the fitted lognormal model.

Quantity	Value
Log-space mean μ	1.43251
Log-space SD σ	0.25091
Median e^μ	4.189×
Multiplicative 1-log-SD factor e^σ	1.285×
One-log-SD band	3.260–5.384×
Two-log-SD band	2.536–6.919×
Linear-scale mean	4.323×
Linear-scale SD	1.102

Table 1: Parameters of the elicited anatomy-factor distribution. For a lognormal distribution, multiplicative log-space bands are more informative than a symmetric linear $\pm$ standard deviation.

6.3 Mesozoic-other scaling

The previous visualization placed non-dinosaurian Mesozoic reptiles halfway between the baseline and dinosaur correction in log space. We preserve that convention by defining

$$F_{\text{meso}} = F_{\text{anat}}^{1/2}.$$

Hence F_{meso} is itself lognormal with median $\sqrt{4.189} = 2.047$ and log-space SD $0.25091/2 = 0.12545$. This remains a visualization heuristic rather than a biomechanical estimate.

6.4 How uncertainty propagates

Conditional on a fixed raw BT score S_i , a dinosaur’s corrected score is $S_i F_{\text{anat}}$. Its median is therefore $4.189S_i$, and its one-log-SD interval is obtained by dividing and multiplying the median by 1.285. For Mesozoic-other entries, the corresponding one-log-SD factor is $e^{\sigma/2} = 1.134$. These bands represent calibration uncertainty only; they do not include uncertainty in the BT fit, crowd composition, body-mass estimates, taxonomy, or graph augmentation.

7 Median-Calibrated Ranking

Table 2 gives representative scores under the median anatomy correction. The ranking itself is constructed from the medians; the uncertainty column shows one log-space standard deviation of the calibration model.

Animal	Crowd	Median corrected	1-log-SD band	Median factor
<i>Tyrannosaurus rex</i>	10,586	44,348	34,506–56,996	4.19×
<i>Utahraptor ostrommaysorum</i>	267	1,118	870–1,436	4.19×
<i>Dakotaraptor steini</i>	191	801	624–1,030	4.19×
<i>Deinonychus antirrhopus</i>	30.1	126	98.1–162	4.19×
<i>Velociraptor mongoliensis</i>	1.66	6.94	5.40–8.92	4.19×
<i>Mosasaurus hoffmannii</i>	5,527	11,311	9,978–12,823	2.05×
Lion	100	100	–	1.00×
Tiger	148	148	–	1.00×
Human	5.2	5.2	–	1.00×

Table 2: Selected animals under the probabilistic anatomy paradigm. Baseline entries are unchanged. Dinosaur and Mesozoic-other intervals reflect calibration uncertainty only.

A useful sanity check is Velociraptor. The median model raises its crowd score from about 1.66 to about 6.9 rather than catapulting it into large-carnivore territory; its one-log-SD calibration band is roughly 5.4–8.9. The same model nevertheless produces substantial changes at larger dromaeosaurid scales: Deinonychus has median score about 126, Dakotaraptor about 801, and Utahraptor about 1,118. In the 123-animal interleaving of Appendix B, Deinonychus lies close to the Lion/Tiger tier, Dakotaraptor below the largest rhinoceroses and elephants but well above the large bears, and Utahraptor above the African Bush Elephant on the median calibration.

8 Weight–Score Analysis

8.1 Median regression

We fit descriptive power laws $s = Cw^b$ in log-space using median regression,

$$\min_{a,b} \sum_i |\log s_i - a - b \log w_i|.$$

Mammals below 1 kg are excluded because the matchup graph provides weak cross-scale calibration there; the theropod sample begins around 0.4 kg. Predictions below those ranges are extrapolations. No bootstrap interval is reported for the fitted exponents, so the curves remain descriptive rather than precision biological scaling laws.

8.2 Median anatomy-paradigm fit

Multiplying every theropod score by a common factor shifts the fitted log-intercept but leaves the exponent unchanged. Replacing the earlier deterministic 10.14×

$$\text{Theropods: } s_{\text{med}} = 0.227 w^{1.263}, \text{ Mammals } (\geq 1 \text{ kg}): s = 0.060 w^{1.376}. \quad (6)$$

The theropod coefficient carries the same anatomy-factor uncertainty as F_{anat} : its one-log-SD coefficient band is approximately 0.176–0.291. The median theropod/mammal ratio at 1 kg is about 3.78×. As before, that vertical separation is not independently discovered by the regression; it is largely introduced by the anatomy calibration. The regression contributes the different mass exponents, which make the median ratio decline with increasing mass.

Weight (kg)	Theropod median	Theropod 1-SD band	Mammal	Median ratio	Context
0.02	0.002	0.001–0.002	0.00028	5.88×	Mouse-sized
0.1	0.012	0.010–0.016	0.003	4.90×	Rat-sized illustrative mammal
2	0.544	0.424–0.700	0.156	3.50×	Squirrel-sized
15	6.94	5.40–8.91	2.49	2.78×	Fox-sized (<i>Velociraptor</i>)
75	53.0	41.2–68.1	22.8	2.32×	Human-sized (<i>Deinonychus</i>)
250	242	188–311	120	2.03×	Lion-sized (<i>Dakotaraptor</i>)
500	581	452–747	310	1.87×	Bear-sized (<i>Utahraptor</i>)

Table 3: Equal-mass predictions under the median anatomy model. The theropod band is one log-space standard deviation of calibration uncertainty. Rows below 1 kg extrapolate the mammal fit; rows below 0.4 kg also extrapolate the theropod fit.

8.3 Dark Era interpretation

The interactive essay describes the deeper intuition in deliberately vivid terms: our baseline intuitions were formed after the disappearance of the non-avian dinosaur regime, and we may therefore treat a comparatively relaxed Cenozoic ecology as normal rather than historically exceptional [1]. The statistical model here translates only one part of that thesis: the possibility that crowd judgments systematically underweight combat-specialized anatomy relative to body mass.

At the median calibration, a 100 g mammal receives a fitted mammal score of about 0.0025. Comparing that baseline with the theropod regression at 0.4 kg, 3 kg, and 15 kg gives cross-mass score ratios of roughly 28×, 360×, and 2,750×, respectively. Those numbers are model-generated score ratios, not predator–prey mortality probabilities or measurements of selection intensity.

The qualitative background is compatible with evidence for a long nocturnal phase in mammalian evolution and with strong ecological restructuring around the K–Pg boundary [6, 5]. At the same time, Mesozoic mammals were ecologically more diverse than the old picture of uniformly tiny nocturnal insectivores suggests [7]. The present score model therefore does not establish the stronger claim that theropod predation caused the mammalian limbic system, sensory architecture, or later human cognition. Those belong to a separate historical hypothesis requiring independent comparative and fossil evidence.

9 Discussion

The probabilistic formulation is an improvement over a single deterministic multiplier in one specific sense: it makes the uncertainty judgment explicit. The model no longer says “dinosaurs are 10.14× stronger than the crowd thinks.” It says: conditional on the anatomy-paradigm hypothesis and the chosen elicitation, the correction factor has median 4.19× and log-space SD 0.251. Readers can therefore see how much of the ranking is stable under moderate movement in the calibration.

The statistical appearance of Eq. (5) should not be confused with data-driven inference. The distribution is *elicited*, not estimated from independent dinosaur combat outcomes. Its center reflects a two-to-one modeling judgment toward the more conservative calibration, and its dispersion is chosen by assigning 15% total mass outside the two elicitation anchors. In Bayesian language it is closer to a prior or sensitivity distribution than to a posterior. Its value is transparency: the subjective step is now parameterized rather than hidden.

The paper’s broader “paradigm” thesis is likewise interpretive. Modern observers have never inhabited a non-avian-dinosaur ecosystem, so there is a plausible psychological route by which body mass becomes an overly dominant intuitive proxy. But the crowd data do not by themselves identify such a cognitive bias. A genuine empirical test would require independent anatomical predictors or performance outcomes, held-out validation, sensitivity to preprocessing choices, and uncertainty in body-mass and taxonomic assignments.

What survives these caveats is a useful exploratory object. The augmented BT graph yields a reproducible crowd baseline; the lognormal correction defines a graded family of anatomy paradigms; and the ranking can be recomputed under any quantile of that distribution. The model therefore converts “dinosaurs may be

systematically underrated” from an all-or-nothing rhetorical claim into an explicit parameter whose consequences can be inspected and eventually tested.

10 Code and Data Availability

The underlying BT pipeline, source data, and interactive visualizations are available at

<https://github.com/brunchwnewton/aethic-code/tree/main/nexic>

and the public interactive essay is at <https://aethus.org/aethic-code/nexic/dino-correct.html>. The present revision analytically replaces the earlier deterministic dinosaur multiplier by Eq. (5); any public visualization still displaying an older fixed multiplier should therefore be read as a historical implementation until synchronized with this paper.

A Theropod Weight Estimates Under the Median Anatomy Model

The following scores are the previous exported theropod values algebraically rescaled from the 10.14 $\times$ scenario to the median 4.189 $\times$ scenario. The one-log-SD column reflects anatomy-factor uncertainty only.

Theropod	Clade	Weight (kg)	Median score	1-log-SD band
<i>Yi qi</i>	Scansoriopterygid	0.4	0.537	0.418–0.690
<i>Archaeopteryx lithographica</i>	Basal avialan	0.9	0.372	0.289–0.478
<i>Compsognathus longipes</i>	Compsognathid	3	0.372	0.289–0.478
<i>Ornitholestes hermanni</i>	Maniraptoran	12	1.16	0.900–1.49
<i>Linheraptor exquisitus</i>	Dromaeosaurid	12	1.61	1.25–2.07
<i>Velociraptor mongoliensis</i>	Dromaeosaurid	15	6.94	5.40–8.92
<i>Dromaeosaurus albertensis</i>	Dromaeosaurid	15	6.78	5.27–8.71
<i>Coelophysis bauri</i>	Coelophysid	20	2.07	1.61–2.65
<i>Suskityrannus hazelae</i>	Tyrannosauroid	30	73.6	57.3–94.6
<i>Stenonychosaurus inequalis</i>	Troodontid	50	18.8	14.6–24.2
<i>Deinonychus antirrhopus</i>	Dromaeosaurid	73	126	98.1–162
<i>Citipati osmolskae</i>	Oviraptorid	75	5.41	4.21–6.96
<i>Moros intrepidus</i>	Tyrannosauroid	78	36.4	28.4–46.8
<i>Liliensternus liliensterni</i>	Coelophysoid	130	45.6	35.5–58.6
<i>Timurlengia euotica</i>	Tyrannosauroid	170	149	116–191
<i>Ornithomimus edmontonicus</i>	Ornithomimid	170	5.16	4.02–6.64
<i>Marshosaurus bicentesimus</i>	Megalosauroid	200	363	282–466
<i>Herrerasaurus ischigualastensis</i>	Herrerasaurid	210	1,045	813–1,343
<i>Dakotaraptor steini</i>	Dromaeosaurid	250	801	624–1,030
<i>Achillobator giganticus</i>	Dromaeosaurid	250	528	411–678
<i>Dilophosaurus wetherilli</i>	Dilophosaurid	400	490	381–630
<i>Gallimimus bullatus</i>	Ornithomimid	440	273	213–351
<i>Utahraptor ostrommaysorum</i>	Dromaeosaurid	500	1,118	870–1,436

Table 4: Theropod points used in the weight–score regression after median recalibration.

B Interleaved Median Rankings: Mammals and Dinosaurs

Table 5 re-ranks the 123-animal selection from the earlier export using the median anatomy factor $4.189\times$ for dinosaurs and median Mesozoic-other factor $2.047\times$. Dinosaurs are marked ▲ and Mesozoic non-dinosaurs △; unmarked entries are baseline. Because the source table was itself rounded for display, very close ranks should be regarded as approximate. Every dinosaur score has the same multiplicative calibration uncertainty $\times/\div 1.285$ at one log-space standard deviation; Mesozoic-other scores use $\times/\div 1.134$.

#	Median score	Animal	Weight	
1	47,289	<i>Ankylosaurus magniventris</i>	6.0 t	▲
2	44,348	<i>Tyrannosaurus rex</i>	8.4 t	▲
3	35,869	<i>Giganotosaurus carolinii</i>	6.5 t	▲
4	33,649	<i>Triceratops horridus</i>	9.0 t	▲
5	22,692	<i>Diplodocus carnegii</i>	13 t	▲
6	14,463	<i>Euoplocephalus tutus</i>	2.5 t	▲
7	13,648	<i>Brontosaurus excelsus</i>	15 t	▲
8	12,247	<i>Spinosaurus aegyptiacus</i>	7.0 t	▲
9	11,719	<i>Stegosaurus stenops</i>	3.5 t	▲
10	11,311	<i>Mosasaurus hoffmannii</i>	5.0 t	△
11	9,180	<i>Daspletosaurus torosus</i>	2.5 t	▲
12	8,012	<i>Styracosaurus albertensis</i>	2.7 t	▲
13	7,141	<i>Tarbosaurus bataar</i>	5.0 t	▲
14	6,141	<i>Allosaurus fragilis</i>	2.0 t	▲
15	3,109	<i>Parasaurolophus spp.</i>	2.5 t	▲
16	2,981	Columbian Mammoth	10.0 t	
17	2,205	Orca (Killer Whale)	5.4 t	
18	1,737	<i>Iguanodon bernissartensis</i>	3.5 t	▲
19	1,609	<i>Edmontosaurus annectens</i>	4 t	▲
20	1,604	<i>Carnotaurus sastrei</i>	1.5 t	▲
21	1,447	Woolly Mammoth	6.0 t	
22	1,446	Woolly Rhinoceros	2.5 t	
23	1,274	<i>Megatherium americanum</i>	4.0 t	
24	1,118	<i>Utahraptor ostrommaysorum</i>	500 kg	▲
25	1,045	<i>Herrerasaurus ischigualastensis</i>	210 kg	▲
26	985	African Bush Elephant	6.0 t	
27	882	White Rhinoceros	2.3 t	
28	853	Great White Shark	1.1 t	
29	801	<i>Dakotaraptor steini</i>	250 kg	▲
30	528	<i>Achillobator giganticus</i>	250 kg	▲
31	490	<i>Dilophosaurus wetherilli</i>	400 kg	▲
32	459	<i>Arctodus simus</i>	800 kg	
33	430	Hippopotamus	1.5 t	
34	417	Saltwater Crocodile	450 kg	
35	363	<i>Marshosaurus bicentesimus</i>	200 kg	▲
36	305	Polar Bear	450 kg	
37	293	<i>Pachycephalosaurus wyomingensis</i>	450 kg	▲
38	274	<i>Smilodon populator</i>	400 kg	
39	273	<i>Gallimimus bullatus</i>	440 kg	▲
40	235	American Bison	900 kg	
41	225	<i>Anzu wyliei</i>	200 kg	▲
42	212	Kodiak Bear	600 kg	
43	203	<i>Smilodon fatalis</i>	220 kg	
44	176	(Wild) Water Buffalo	900 kg	

#	Median score	Animal	Weight	
45	161	Water Buffalo	800 kg	
46	160	Wood Bison	900 kg	
47	150	American Alligator	230 kg	
48	149	<i>Timurlengia euotica</i>	170 kg	▲
49	148	Tiger	220 kg	
50	144	<i>Quetzalcoatlus northropi</i>	250 kg	△
51	133	Grizzly Bear	270 kg	
52	126	<i>Deinonychus antirrhopus</i>	73 kg	▲
53	109	Alaskan Moose	600 kg	
54	100	Lion	190 kg	
55	88.0	Siberian Tiger	230 kg	
56	85.0	African Lion	185 kg	
57	81.0	Bengal Tiger	220 kg	
58	73.5	<i>Suskityrannus hazelae</i>	30 kg	▲
59	48.0	Jaguar	96 kg	
60	45.4	<i>Liliensternus lilienseni</i>	130 kg	▲
61	37.0	Western Gorilla	160 kg	
62	36.4	<i>Moros intrepidus</i>	78 kg	▲
63	31.0	Cougar	70 kg	
64	30.0	Leopard	60 kg	
65	24.0	Eastern Gorilla	160 kg	
66	23.0	Komodo Dragon	70 kg	
67	20.0	Green Anaconda	30 kg	
68	19.0	<i>Stenonychosaurus inequalis</i>	50 kg	▲
69	18.0	Wild Boar	80 kg	
70	17.0	Spotted Hyena	55 kg	
71	17.0	Dire Wolf	68 kg	
72	15.7	<i>Protoceratops andrewsi</i>	180 kg	▲
73	13.0	Elk	330 kg	
74	10.0	Common Warthog	80 kg	
75	9.7	Grey Wolf	40 kg	
76	7.9	Burmese Python	90 kg	
77	7.5	Neanderthal	78 kg	
78	7.0	<i>Velociraptor mongoliensis</i>	15 kg	▲
79	6.6	<i>Dromaeosaurus albertensis</i>	15 kg	▲
80	6.4	Bornean Orangutan	75 kg	
81	5.9	Human (male)	80 kg	
82	5.9	Neapolitan Mastiff	60 kg	
83	5.8	Ostrich	110 kg	
84	5.7	Dogo Argentino	43 kg	
85	5.4	<i>Citipati osmolskae</i>	75 kg	▲
86	5.4	<i>Ornithomimus edmontonicus</i>	170 kg	▲
87	5.2	Human	70 kg	
88	4.8	Caucasian Shepherd Dog	55 kg	
89	4.5	Rottweiler	50 kg	
90	4.4	Cheetah	50 kg	
91	4.3	Tibetan Mastiff	55 kg	
92	4.2	Red Kangaroo	85 kg	
93	3.9	Wolverine	14 kg	
94	3.5	Mandrill	25 kg	
95	3.3	Common Chimpanzee	50 kg	
96	3.1	Kangal	55 kg	
97	2.9	Southern Cassowary	60 kg	

#	Median score	Animal	Weight	
98	2.9	<i>Pteranodon longiceps</i>	25 kg	△
99	2.5	Boerboel	65 kg	
100	2.4	<i>Psittacosaurus lujiatunensis</i>	20 kg	▲
101	2.2	Thylacine	25 kg	
102	2.1	German Shepherd	34 kg	
103	2.1	<i>Coelophysis bauri</i>	20 kg	▲
104	1.8	Human (female)	55 kg	
105	1.8	Ocelot	12 kg	
106	1.7	King Cobra	6 kg	
107	1.7	American Pit Bull Terrier	27 kg	
108	1.7	Honey Badger	11 kg	
109	1.6	<i>Linheraptor exquisitus</i>	12 kg	▲
110	1.5	Bonobo (Pygmy Chimpanzee)	40 kg	
111	1.5	Caracal	13 kg	
112	1.4	Fishing Cat	10 kg	
113	1.3	Bobcat	9 kg	
114	1.2	Coyote	14 kg	
115	1.2	<i>Ornitholestes hermanni</i>	12 kg	▲
116	1.1	Labrador Retriever	30 kg	
117	1.0	Savannah Cat	8 kg	
118	0.54	<i>Yi qi</i>	400 g	▲
119	0.50	Brown (Norway) Rat	300 g	
120	0.37	<i>Archaeopteryx lithographica</i>	900 g	▲
121	0.37	<i>Compsognathus longipes</i>	3 kg	▲
122	0.10	House Mouse	20 g	
123	0.10	Etruscan Shrew	2 g	

Table 5: Interleaved ranking under the median probabilistic anatomy paradigm (Lion = 100). The table is ordered by median score; calibration uncertainty is described in the text rather than repeated row by row.

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